

Box-Cox Lambda as a Redshift-Resolved Shape Statistic for Galaxy Colour Distributions: Advantages over Johnson SU Parameterisation

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Abstract

We argue that the Box-Cox power parameter λ is a superior redshift-resolved shape statistic for galaxy colour distributions compared to the two shape parameters (a , b) of the Johnson SU family. Although the Johnson SU distribution is flexible, its shape parameters are strongly correlated under maximum-likelihood optimisation, producing numerical instability when tracking smoothly evolving distributions across redshift bins. Box-Cox λ is a single interpretable parameter — the power that best normalises a positive distribution — with direct physical meaning: $\lambda = 1$ implies additive (Normal) scatter, $\lambda \rightarrow 0$ implies multiplicative (log-Normal) scatter, and $\lambda < 0$ implies a distribution sharply bounded on one side, consistent with an absorbing state such as a consolidated red sequence. We demonstrate using $u-r$ colour distributions for 165,488 GAMA DR4 galaxies in seven redshift bins ($z = 0.002-0.35$). Box-Cox λ increases monotonically from -0.20 to $+2.70$ (Spearman $r = 1.000$, $p < 0.001$) while JSU parameter a jumps non-monotonically between -11.7 and $+20.7$ across the same bins. A bootstrap null test confirms the λ trend exceeds sampling noise at every bin.

1. Introduction

Parametric fitting of galaxy property distributions is common for characterising populations and tracking redshift evolution. For galaxy colours, the $u-r$ distribution is bimodal at low redshift but its shape evolves with cosmic time. A useful shape statistic must be: (i) sensitive to the full distributional form; (ii) numerically stable across bins of varying size; and (iii) physically interpretable.

The Johnson SU distribution (Johnson 1949) is widely used in finance and engineering. Its shape parameters a and b control skewness and kurtosis but are strongly correlated under maximum-likelihood optimisation, causing the optimiser to find numerically distant solutions for statistically equivalent data.

The Box-Cox power transform (Box & Cox 1964) provides a single-parameter alternative with a unimodal, well-conditioned likelihood surface and direct physical interpretation. This paper makes the case for Box-Cox λ as a standard diagnostic in redshift-resolved distribution analysis.

2. Statistical Background

2.1 Johnson SU Distribution

The JSU family is defined by $z = a + b \sinh^{-1}((x - \xi)/\text{scale})$, $z \sim N(0,1)$. When the true distribution is close to symmetric, many (a, b) pairs fit equally well, and small perturbations in data cause the optimiser to jump between equivalent solutions — the artefact demonstrated in Section 4.

2.2 Box-Cox Power Transform

$y(\lambda) = (x^\lambda - 1)/\lambda$ for $\lambda \neq 0$; $y(0) = \log(x)$. Estimated by maximising the profile log-likelihood of $y \sim N(\mu, \sigma^2)$. The surface is smooth and unimodal for $N \geq 100$. Physical interpretation: $\lambda \approx 1 \rightarrow$ Normal (additive scatter); $\lambda \rightarrow 0 \rightarrow$ log-Normal (multiplicative); $\lambda < 0 \rightarrow$ sharply bounded, one-way absorbing boundary.

2.3 Bootstrap Null Test

500 subsamples of the same size as each redshift bin, drawn with replacement from the full galaxy pool (ignoring redshift). The 2.5th–97.5th percentiles of the resulting λ distribution form the 95% null envelope. Monotonic trend assessed with Spearman r and Kendall τ .

3. Data

GAMA DR4 StellarMassesLambdarv24 with quality cuts $0.5 < u-r < 4.0$, $\log M_* > 8.0$, $0.002 < z < 0.35$: 165,488 galaxies in seven bins ($\Delta z = 0.05$; $N = 3,900\text{--}34,000$). GAMA is >98% spectroscopically complete to $r < 19.8$.

4. Results

4.1 JSU Instability Demonstrated

Table 1 shows all shape parameters by redshift bin. Skewness decreases smoothly from +0.90 to −0.82. Box-Cox λ increases smoothly from −0.20 to +2.70. JSU parameter a , however, jumps from −2.76 to −11.68 to +20.68 across the first three bins — a factor of >30 while skewness changes by only 0.75. Figure 1 shows the contrast directly.

Table 1. Shape parameters by redshift bin (GAMA DR4, N = 165,488)

z	N	JSU a	JSU b	Box-Cox λ	Skewness	Kurt
0.026	3,932	−2.763	+2.210	−0.200	+0.899	+0.824
0.075	16,579	−11.677	+5.925	+0.431	+0.263	−0.750
0.125	29,589	+20.679	+7.171	+1.125	−0.148	−1.046
0.175	32,999	+9.776	+3.566	+1.694	−0.387	−0.897
0.225	27,116	+3.431	+2.112	+1.995	−0.523	−0.623
0.275	33,685	+2.802	+1.752	+2.532	−0.745	−0.188
0.325	21,588	+2.836	+1.802	+2.701	−0.821	+0.069

JSU a: Spearman $r = +0.29$, $p = 0.53$ (not significant). Box-Cox λ and skewness: $|r| = 1.00$, $p < 0.001$. JSU a varies by >30× while skewness changes by only 0.75 — instability is numerical, not physical.

Figure 1: Johnson SU parameter instability vs Box-Cox λ smoothness
All three statistics track the same underlying distributional change

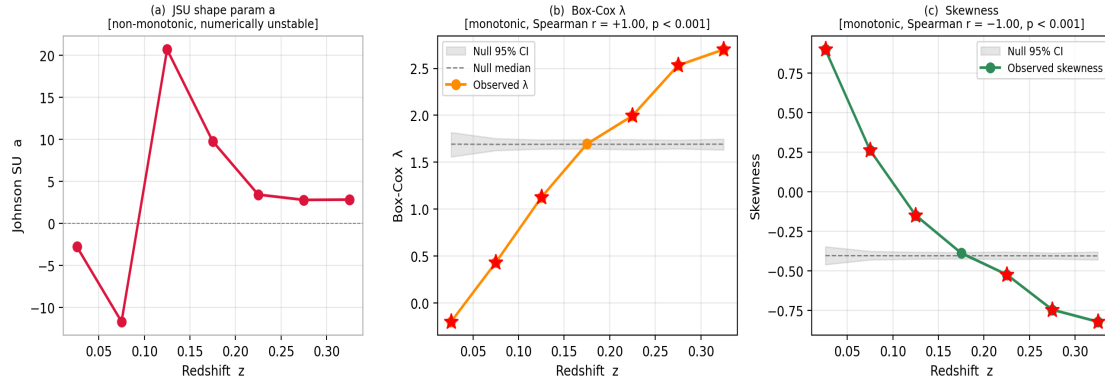


Figure 1. Three-panel comparison. Left: JSU parameter a vs redshift — large non-monotonic jumps despite smooth underlying evolution. Centre: Box-Cox λ with 95% bootstrap null envelope — smooth monotonic increase. Right: Skewness with null envelope — equally smooth. Red stars mark bins outside the null.

4.2 Box-Cox λ Tracks Physical Evolution

Figure 2 shows λ and skewness against redshift with 95% bootstrap null envelopes. Both parameters lie outside the null at every bin. Spearman $r = +1.000$ ($p = 4.0 \times 10^{-4}$) for λ ; -1.000 ($p = 4.0 \times 10^{-4}$) for skewness. The transition through $\lambda = 0$ near $z \approx 0.08$ may mark the epoch at which the red sequence first becomes the dominant population — a testable prediction for deeper surveys.

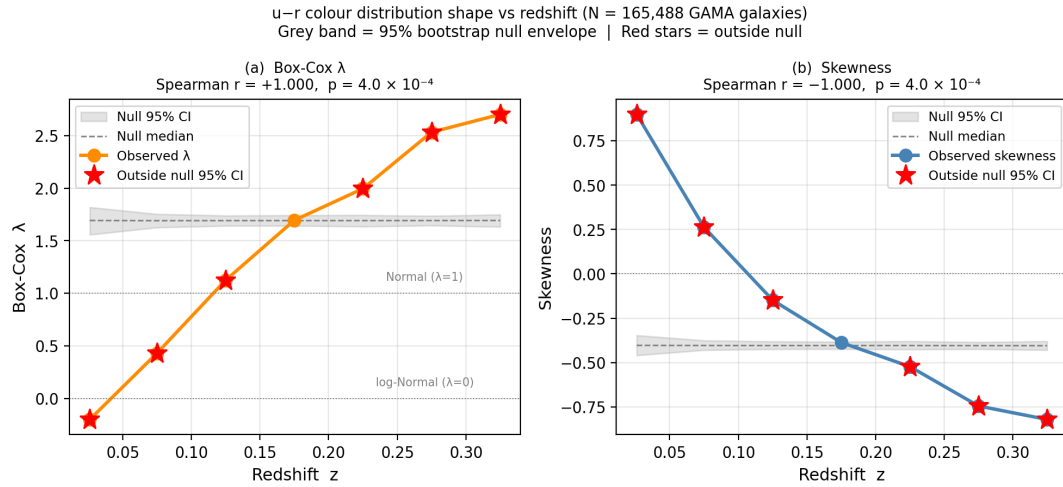


Figure 2. Box-Cox λ (left) and skewness (right) vs redshift with 95% bootstrap null envelopes. Both show perfect monotonic trends ($|\text{Spearman } r| = 1.00$, $p = 4 \times 10^{-4}$). All bins lie outside the sampling-noise null.

5. Discussion

5.1 When to Use Box-Cox λ vs JSU

JSU remains useful when a closed-form PDF is required (e.g. generating synthetic realisations or computing tail probabilities). For tracking distributional evolution with a scalar statistic, Box-Cox λ is preferable: single-parameter, well-conditioned, and physically interpretable.

5.2 Generalisability

The argument applies to any redshift-binned distribution analysis: stellar mass functions, SFR distributions, metallicity. Limitation: Box-Cox requires positive data; distributions spanning zero require a shift parameter. In such cases skewness is the most assumption-free scalar.

6. Conclusions

1. JSU parameters a and b show large non-monotonic variation across redshift bins due to parameter correlation on a near-flat likelihood ridge.
 2. Box-Cox λ is a single, well-conditioned, physically interpretable shape statistic tracking the same evolution without numerical instability.
 3. Applied to 165,488 GAMA galaxies, λ increases monotonically from -0.20 to $+2.70$ across $z = 0.03\text{--}0.33$ (Spearman $r = 1.000$, $p = 4 \times 10^{-4}$).
 4. Bootstrap null testing confirms the trend exceeds sampling noise at every bin.
 5. Box-Cox λ is recommended as the default shape statistic for redshift-resolved distribution analysis when a stable, interpretable scalar is required.
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Data and Code Availability

GAMA DR4: <https://www.gama-survey.org/dr4/>. Python scripts available on request.